

Privacy-preserved Intermediate Feature Compression for Cyber-Physical Systems

Zixi Wang

Xi'an Jiaotong University

Xi'an, China

wzx284788391@stu.xjtu.edu.cn

Le Yang

Xi'an Jiaotong University

Xi'an, China

yangle15@xjtu.edu.cn

Yuan Zhang

China Telecom Research Institute

Shanghai, China

zhangy666@chinatelecom.cn

Fan Li

Xi'an Jiaotong University

Xi'an, China

lifan@mail.xjtu.edu.cn

ABSTRACT

As an important component of Cyber-Physical Systems (CPS), wireless sensor network faces challenges of computation, communication and privacy. For employing DNN with large capacity in CPS, splitting computing (SC) is proposed, in which transmitting intermediate data is the bottleneck of system efficiency. In this paper, targeting to the compression and privacy limitations occur in the original extracted feature data, we proposed a privacy-preserved intermediate feature compression framework. Therein, a hybrid feature compression module composed of a knowledge-based adaptor network and a traditional video codec is proposed. A noise-based privacy enhancement and domain-concentration cooperative privacy preserving approach is developed. Evaluations are conducted on two DNN models among two datasets, proving the proposed method can achieve better analysis accuracy as well as higher degree of privacy preserving than HEVC.

CCS CONCEPTS

• **Human-centered computing** → Ubiquitous and mobile computing; • **Computer systems organization** → Embedded and cyber-physical systems; • **Security and privacy** → Systems security.

KEYWORDS

feature compression; privacy preserving; cyber-physical systems; deep neural networks

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1 INTRODUCTION

Cyber-Physical Systems (CPS) [20] are designed to integrate the networking, computation technologies, and physical systems. The advancements of CPS have been verified in various domains, such as industrial, health-care, transportation and manufacturing monitoring systems [2–4, 35, 38, 39]. Wireless sensor network and distributed computing are important parts of the implementation of CPS [37], to integrate and process physical data to handle the monitoring task. In the context of deep neural network (DNN) based data processing technology, a common approach is splitting computing (SC) [24, 36, 40], i.e., to split a well-pretrained DNN model into two parts and deploy them to the front sensors and the back-end server respectively. To maintain efficient application of CPS, adopting SC framework faces three challenges, i.e., compression efficiency, module portability and privacy.

Compression efficiency challenge mostly stems from the transmitting process of intermediate feature. In SC, the intermediate feature generated by the former part of network will be transmitted from sensors to the server, and performs as the input of the later network. Therefore, the volume of transmitted data directly determines the transmitting time when channel bandwidth is limited. An analysis-oriented DNN model will normally not take compression performance into consideration, thus directly transferring the raw intermediate feature will be time-consuming and effect on the efficiency of the whole framework. Therefore, exploring efficient feature compression method is necessary.

In this context, module portability also becomes an essential property to satisfy. Some studies introduce learning-based feature compression methods [12, 17, 23, 28] and using end-to-end optimization to fine-tune the model parameters of original DNN as well as train the compression module collaboratively. Despite of competitive compression performance in these studies, adopting parameter modification for the given well-pretrained DNN model is not practical in CPS, because such modification will involve all physical sensors in the system and require lots of training samples. Also, the compression module and the modified DNN model must be coupled with those approaches, implying that they are not portable. To this end, there are some researches investigating the compression efficiency of feature with no model parameter modification, by improved methods based on video codec [5–10, 16, 30].

Meanwhile, privacy challenge will stand when users only want the server to accomplish data analysis task, but do not want their

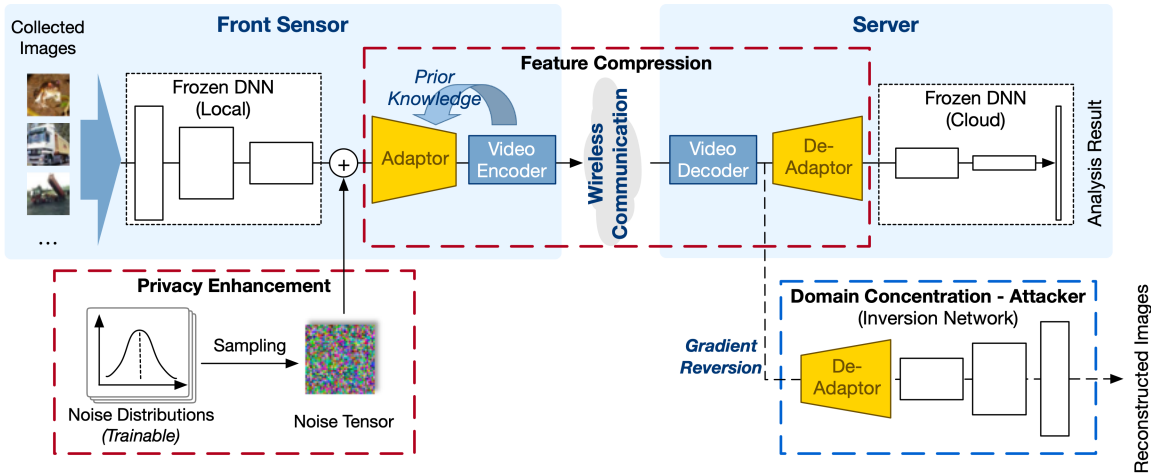


Figure 1: The framework of privacy-preserved intermediate feature compression. The arrows with solid line indicate the workflow in inference process, whereas the arrow with dashed line indicate that this branch is only for training process.

personal information exposed. Transferring intermediate feature rather than raw physical data to the cloud end shall protect privacy in some degrees, however, there is still potential information containing in the intermediate feature that may lead to privacy leakage with model inversion attack [1, 33]. Targeting to protect privacy in SC applications, researches have been studied in terms of differential privacy [18, 21, 22, 25, 26, 34, 41], in which a common approach is to add noise tensor to the intermediate feature. Whereas from the sight of compression, addition noise will introduce extra uncertainty and chaos to the data space, which may increase the information entropy as well as the compressed bit-rate [27]. To this end, the trade-off between privacy-preserving and feature compression needs to be further explored.

In this paper, we take compression efficiency, module portability and privacy into consideration simultaneously, and proposed a novel private intermediate feature compression framework for image analysis tasks in CPS with specified data analysis DNN model. Better analysis accuracy as well as higher degree of privacy are demonstrated according to experiments on two DNN models among two datasets.

- A novel compression framework that emerges privacy preserving as well as feature compression and can be adapted to different backbone networks is proposed. There is no need to conduct any structure modification or parameter fine-tuning for the given backbone. By optimizing a portable module that is concatenated serially after the extracted feature, not only the local private information can be protected but high compression ratio for intermediate feature can be achieved.
- A noise-based privacy enhancement and domain-concentration cooperative privacy preserving method is proposed. A trainable noise tensor is added to the extracted intermediate feature to directly disturb its private information. During training process, an attacker network is adopted to the intermediate feature to help concentrating its distribution domain to the analysis task and restraining the other private knowledge, where the measurement of privacy is formulated according

to the knowledge of information theory. A gradient-reversal based training strategy is proposed to improving privacy as well as maintaining analysis accuracy.

- A portable hybrid feature compression module composed of a knowledge-based adaptor network and a video codec is proposed. For a video codec, a prior knowledge goes that the stronger internal correlations the input data possess, the better compression efficiency the codec can achieve. Therefore, the specifically designed adaptor network optimizes the distribution of the perturbed feature after noise addition and rearrange it, for compression demand. Different compression ratios can be achieved by adjusting the coding parameters of video codec rather than different parameter sets of adaptor network, saving training costs and storage resources.

2 PRIVACY-PRESERVED INTERMEDIATE FEATURE COMPRESSION

Fig. 1 illustrates the overall framework of the proposed method. In general, three additional modules, i.e., feature compression, privacy enhancement and domain concentration, are specifically designed. With the specified DNN backbone for image analysis, a noise tensor sampled by a set of learned noise distributions will be added to the extracted intermediate feature, for privacy preserving. Then a knowledge-based adaptor network is conducted to modified the perturbed feature to improve the coding efficiency that the subsequent video codec could achieve. During the training process, an inversion network performs as an attacker to simulate the potential privacy attacks. With gradient-reversal based training strategy, the distribution domain of the transmitted feature can be concentrated on analysis task whereas the privacy knowledge can be restrained.

2.1 Channel-Wise Trainable Noise-based Privacy Enhancement

In the sight of intermediate feature, a prior knowledge goes that each feature channel's tolerance of noise addition can be quite

different, when the accuracy losses are same. It is intuitive, because different channel has different importance to the final analysis. In order to optimize the trade-off between privacy and analysis, we propose a channel-wise trainable noise addition method.

Given a group of intermediate feature tensor F with size of (M, N, C) , we model C different normally noise distributions $N(\mu_c, \sigma_c^2)$ respectively to formulate the noise tensor for each feature channel, where $c = \{1, 2, \dots, C\}$. To enable gradient-based training on μ_c and σ_c , the sampled noise tensor $n_c \sim N(\mu_c, \sigma_c^2)$ is calculated by $n_c = \sigma_c \cdot e_c + \mu_c$, where $e_c \sim N(0, 1)$. Note that σ_c should be positive value following its physical meaning, so re-parameterization $\sigma_c = \frac{1.0 + \tanh(\rho_c)}{2} \cdot T$ is developed to fit the limitation during training process. In this formulation, ρ_c is defined as a trainable parameter in gradient descent optimization, and the value range of ρ_c is unlimited whereas σ_c can be constrained to a positive value lower than a given upper bound T . In this way, two trainable parameters μ_c and ρ_c are introduced to each feature channel, then total $2 \times C$ additional parameters are involved in noise addition module. The detailed training strategy will be explained in Section 2.3. The perturbed feature F_p is then calculated by $F_p = F + n$, where n is a tensor concatenated by $\{n_1, n_2, \dots, n_C\}$ and is of the same size as F .

2.2 Hybrid Feature Compression with Knowledge-based Adaptor

After noise addition, the perturbed feature will be more disorder than the initial feature. Therefore, direct adopting traditional video codec to compress such perturbed feature can be low efficient. To this end, an adaptor network is developed firstly before the feature is fed into video codec. When server receives the compressed data, video decoder is conduct to decompress it, then a de-adaptor network is developed to resume the feature distribution for satisfying the later part of backbone DNN. Note that video codec is non-differential and can't be involved in gradient-based training, so the prior knowledge that coding efficiency is directly relevant to the input data's internal correlation is used when designing adaptor network, and the goal of adaptor network is to rearrange the perturbed feature and manifest its internal correlation just as the pixel correlation shows in video data.

The detailed network architecture of adaptor and de-adaptor are as depicted in Fig. 2. There are only four convolutions with kernel size of 3 consisting in the adaptor network, which guarantees that such module is lightweight. The channel size is omitted in this illustration, because this parameter will be determined by the channel dimension C of the to-be-compressed feature $F_p \in R^{M \times N \times C}$. Specifically, the output channel of the first convolution is same

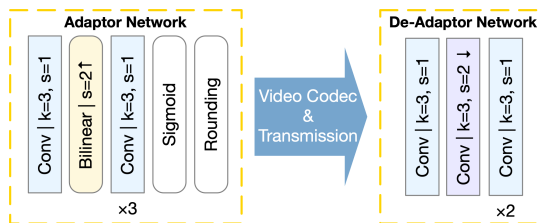


Figure 2: Network architecture of adaptor and de-adaptor.

to C , whereas the three convolutions after bilinear up-scaling is $\frac{C}{4}$. By this design, the size of adapted feature F_a is $(2M, 2N, \frac{C}{4})$, so the total number of tensor element is unchanged and no volume loss will be introduced in this process. Sigmoid activation and rounding operation are conducted at the end of this module, in order to normalize F_a into 8-bits integer format for fitting the input requirement of normal video codec. Implicitly, adaptor network extracts the correlations of each four channels, and rearrange their elements into a larger feature map to present such correlations explicitly. At the server end, de-adaptor network is designed as a mirror of adaptor network, where the only difference is that the bilinear up-scaling is replaced by a convolution with stride of 2.

For helping video codec further exploit the inter-channel correlations of F_a , the feature maps are tiled into one frame by descending order of their element's mean. Then video codec can compress it by INTRA mode and make full use of the spatial correlations.

2.3 Gradient-Reversal based Domain Concentration

The noise addition method in Section 2.1 can reduce intermediate feature's signal-to-noise ratio in a straightforward way, but a more concrete optimization objective is necessary to improve privacy preserving efficiency by determining a quantitative index to measure privacy.

From the knowledge of information theory [31], we formulate the target of privacy preserving is to minimizing the mutual information consisting in the compressed data and the original input. However, the goal of analysis task requests maximizing the mutual information between the intermediate feature and the real label [32]. Therefore, the essential optimization objective is to concentrate feature domain on the analysis task whereas restrain the other information related to input data. Our total loss function can then be formulated as

$$L = -\alpha \cdot \log \frac{1}{C} \sum_{c=0}^C \sigma_c^2 + I(F_a, Y) + R_{-\lambda}(I(F_a, X)). \quad (1)$$

In this formulation, the first term is the mean deviation of learned noise tensor, where α is a weight parameter. The goal of this term is straightforward to maximize the amplitude of noise addition, so signal-to-noise ratio can be reduced. $I(F_a, Y)$ and $I(F_a, X)$ denote the mutual information between F_a and real label Y as well as input X , respectively. It is normally hard to calculate the concrete value of mutual information for multi-dimension data. Therefore, in our implementation, $I(F_a, Y)$ is implied by normal objective function in analysis task, such as cross entropy in classification task. An inversion network referring to the former part of backbone DNN is developed as an attacker. Then the similarity between reconstructed image and input reflects $I(F_a, X)$. Gradient reversion operation [14] is introduced as $R_{-\lambda}(x)$, defining as

$$\begin{aligned} R_{-\lambda}(x) &= x, \\ \frac{dR_{-\lambda}}{dx} &= -\lambda. \end{aligned} \quad (2)$$

In this way, the attacker module itself is optimized for maximizing the quality of reconstructed image (peak signal to noise ratio,

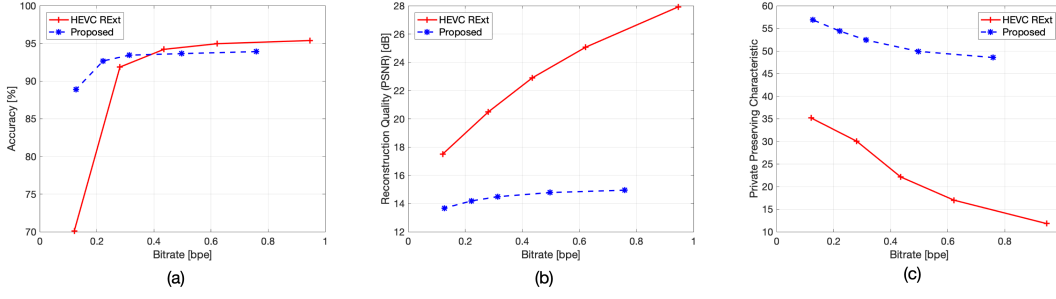


Figure 3: Comparison performance on CIFAR-10 when using HEVC RExt and our proposed method in terms of (a) RA curves, (b) RQ curves and (c) PPC curves.

PSNR is used in our experiments), whereas from the sight of former parts of the system the target is to minimizing the final PSNR. Therefore, the second and third terms in Equation 1 achieve concrete domain concentration to prevent inversion attack. α and λ should be adjusted collaboratively in experiments for better performance.

3 EVALUATIONS

3.1 Implementation Settings

We use image classification task to verify the proposed method in our experiments. We evaluate two datasets on two DNN models, i.e., ImageNet [11] on ResNet50 [15] and CIFAR-10 [19] on ResNet34. The DNN models are split at the end of second residual block. For ImageNet, the size of input image is $224 \times 224 \times 3$, so the size of intermediate feature tensor extracted by ResNet50 is $28 \times 28 \times 512$. For CIFAR-10 using ResNet34, the two terms are $32 \times 32 \times 3$ and $16 \times 16 \times 128$ respectively. When using different DNN backbones, the differences in the proposed feature compression approach are just the channel dimension C in the adaptor network and the architecture of inversion network in training process.

For end-to-end training of the whole framework, we omit the video codec during training by regarding this process as lossless, which has been proved to have no evident impact on the final performance by experiments. In mini-batch learning, input images are fed into local DNN, adaptor, de-adaptor and cloud DNN sequentially, to generate a prediction label so as to calculate cross entropy loss. The features after adaptor network are also fed to attacker network, to obtain reconstruction images. The parameters of DNN backbone are frozen during back propagation.

For systemically evaluating the proposed method, we conduct evaluations in terms of both compression performance and reconstruction performance, by rate-accuracy (RA) curves and rate-reconstruction quality (RQ) curves. Specifically, the compressed bits per element (bpe), calculated by dividing the size of output bit-stream by the total number of feature tensor elements, is used to measure the compression ratio. Note that for RQ curves, a reinitialized inversion network, but not the one that has been fully-trained before, is trained when all the other parts of the framework are frozen, to simulate the real attacker in inference process and evaluate the quality of reconstructed image. Following [13], privacy-preserving characteristics (PPC) are also evaluated to illustrate the trade-off of those two terms. PPC is measured by the product

of final analysis accuracy and distortion of reconstructed images. Therefore, the higher the PPC value is, the better the performance is at preserving privacy with less analysis accuracy loss.

For comparison, video codec based feature compression approach as presented in [6] is implemented. We use HEVC [29] Range extension (RExt) to conduct the experiment, by official reference software HM16.20.

3.2 Results on CIFAR-10

For CIFAR-10, the model of ResNet34 is firstly reproduced and 95.61% classification accuracy is achieved by our own training. Then the model is frozen as the backbone DNN and split to be employed on front sensors and server respectively. For training the modules involved in our proposed method, initial models of adaptor and de-adaptor are firstly trained by a stochastic gradient descent (SGD) optimizer with momentum of 0.9 and batch size of 128. The weight decay parameter is set to 0.0005. The learning rate is initialized by 0.1 and scheduled by cosine annealing with total 200 epochs. With such initialization, the framework is trained collaboratively according to loss function Eq. 1. Specifically, λ is set to 0.001 and α is set to 0.01 by experiments. Another SGD optimizer is employed, where learning rates for feature compression module, privacy enhancement module and domain concentration module are 0.0003, 0.01 and 0.00001 respectively. For achieving different compression ratio, five different QPs = {32, 37, 42, 45, 49} are set in the video codec used in the proposed method.

The comparison results are illustrated in Fig. 3. In terms of RA curves, the proposed method outperform HEVC RExt when bit-rate is lower than 0.4 bpe. The improvement grows as the compression ratio increases. Specifically at 0.12 bpe, the proposed method

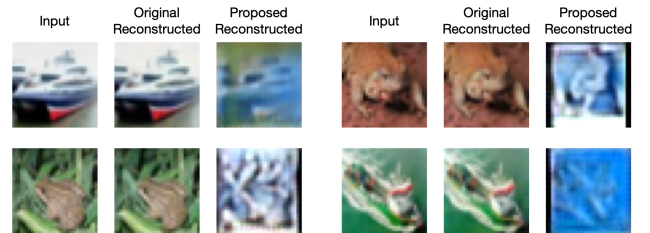


Figure 4: Visualization results of reconstructed images.

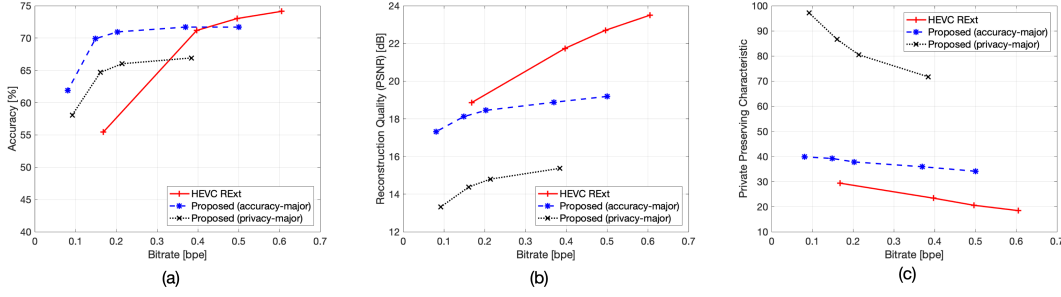


Figure 5: Comparison performance on ImageNet when using HEVC RExt and our proposed method in terms of (a) RA curves, (b) RQ curves and (c) PPC curves. Two models of the proposed method targeting to higher accuracy and higher privacy are presented as “accuracy-major” and “privacy-major”, respectively.

still achieves 88.89% accuracy whereas HEVC RExt only present 70.12%. Such improvements profit from the adaptor network who modifies intermediate feature space for helping video codec’s compression. Note that the proposed method achieves slightly lower accuracy when bit-rate grows higher. This is introduced by the inevitable trade-off caused by privacy preserving. As presented in Fig. 3(b), reconstruction quality of the proposed method is much lower than HEVC RExt at all points of bit-rate. The PSNR reduction even achieves 15dB at high bit-rate. These results indicates that the proposed method evidently reduces the mutual information between intermediate feature and input image, and possesses better privacy preserving ability facing inversion attack. PPC curves presented in Fig. 3(c) reveal the trade-off efficiency between privacy and accuracy. Apparently, PPC values of the proposed method are all higher than HEVC RExt, indicating that the proposed method indeed present better privacy preserving performance as well as lower accuracy loss.

Fig. 4 gives the visualization results of reconstructed images on CIFAR-10, to intuitively illustrate the privacy preserving performance. Images in the first and forth columns are the original input in CIFAR-10 dataset, and those in the second and fifth columns are the reconstructed images generated by the attacker model trained on the original intermediate feature. Images in the third and last columns are generated by the attacker model trained on the perturbed intermediate feature extracted by the proposed method, in which edge and color distortion can be intuitively observed.

3.3 Results on ImageNet

For ImageNet, the ResNet50 pretrained model with top1 accuracy of 76.01% is used as the backbone DNN. Similar to the experiments on CIFAR-10, adaptor and de-adaptor networks are initialized by a SGD optimizer with learning rate of 0.0001, momentum of 0.9 and weight decay of 0.0001. The batchsize is set to 256. Then collaborative trainings according to Eq. 1 are conducted with two settings, $\lambda = 0.1, \alpha = 3$ and $\lambda = 0.5, \alpha = 5$. This is because ImageNet is a larger and more complex dataset composed of 1000 diverse classes so privacy preserving will inevitably introduce more accuracy loss than CIFAR-10. Different settings of λ and α provide more choices of the trade-off for fitting different demands in practical applications. SGD optimizer with learning rates for feature compression module,

privacy enhancement module and domain concentration module of 0.0001, 0.01 and 0.00001 respectively are employed in this process. Five different QPs = {28, 32, 39, 42, 47} are set in the video codec used in feature compression module.

As presented in Fig. 5, the model trained with $\lambda = 0.1, \alpha = 3$ is denoted as “Proposed (accuracy-major)”, whereas the one trained with $\lambda = 0.5, \alpha = 5$ is denoted as “Proposed (privacy-major)”. Intuitively, lower values of λ and α lead to higher accuracy but worse privacy performance, just as (a) and (b) in Fig. 5 show. Specifically, the accuracy-major model achieves more than 70% accuracy at 0.37 bpe, whereas the privacy-major model achieves 66.92% and has nearly reached its upper bound. However, the privacy-major model is able to provide lower reconstruction quality that less than 14dB. The PPC curves also support that those two models actually present different degrees of trade-off between privacy and accuracy. However, despite of employing different settings, the two models of proposed method both outperform HEVC RExt in overall. Especially at bit-rates lower than 0.3, the proposed method always achieve better accuracy and lower reconstruction quality than HEVC RExt, validating ours’ efficiency.

4 CONCLUSIONS

In this paper, a privacy-preserved intermediate feature compression framework is proposed to solve the compression and privacy limitations in original intermediate data generated by SC. For privacy concerns, a channel-wise trainable noise-bases privacy enhancement method is developed to reduce the data’s signal-to-noise ratio, improving the difficulty of conduct inversion attacks. Then a hybrid feature compression method with knowledge-based adaptor network is proposed to improve the coding efficiency of perturbed feature. A gradient-reversal based domain concentration training strategy is further proposed so that the mutual information of input consisting in the intermediate feature can be reduced while maintaining analysis performance. Finally, experiments on two DNN models among two classification datasets validate the efficiency.

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